



AZ-204T00A
Learning Path 01:
Implement Azure App
Service web apps

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These Learn modules are part of the AZ-204: Implement Azure App Service web apps (<https://learn.microsoft.com/training/paths/create-azure-app-service-web-apps/>) learning path.

Agenda

- Explore Azure App Service
- Configure web app settings
- Scale apps in Azure App Service
- Explore Azure App Service deployment slots

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Module 1: Explore Azure App Service

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<https://learn.microsoft.com/training/modules/introduction-to-azure-app-service/>

Learning objectives

- Describe Azure App Service key components and value.
- Explain how Azure App Service manages authentication and authorization.
- Identify methods to control inbound and outbound traffic to your web app.
- Deploy an app to App Service using Azure CLI commands.

Introduction

- Azure App Service is an HTTP-based service for hosting web applications, REST APIs, and mobile back ends.
- Applications run and scale in both Windows and Linux-based environments

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Examine Azure App Service

Built-in scale support

- Save costs and meet demand through scaling
- Scale out/in manually or automated based on metrics
- Scale up/down by changing the app service plan

Continuous integration/ deployment support

- Azure DevOps
- GitHub
- Bitbucket
- FTP
- Local Git repository
- Container registry
- And others

Deployment slots

- Target deployments to test or production environments
- Customize what settings are swapped between slots

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Examine Azure App Service plans

App Service plans

- Define a set of compute resources
- Can run one or more apps in the same plan
- Can be scaled up or down at any time to meet compute or feature needs

Usage tiers

- Shared: shared compute resources targeting dev/test
- Basic: dedicated compute targeting dev/test
- Standard: run production workloads
- Premium: Enhanced performance and scale
- Isolated: high-performance, security and isolation

How does my app run and scale?

- Shared: apps receive CPU minutes on a shared VM instance and can't scale out
- Other tiers: apps run on all the VM instances configured in the App Service plan

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If your app is in the same App Service plan with other apps, you may want to improve the app's performance by isolating the compute resources.

Deploy to App Service

Every development team has unique requirements for their deployment pipeline. App Service supports both automated and manual deployment.

Automated deployment

- Azure DevOps
- GitHub
- Bitbucket

Manual deployment

- Git
- CLI
- Zipdeploy
- FTP/S

Deployment slots

- Target deployments to test or production environments
- Customize what settings are swapped between slots

Sidecar containers

- Add up to nine sidecar containers for each side-car enabled container app.
- Add through the Deployment Center in the app's management page.

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Explore authentication and authorization in App Service

- Built-in authentication and authorization support
 - Implement with low to no code changes in your web app
- Identity providers available by default
 - Microsoft Entra
 - Facebook
 - Google
 - X
 - Apple (Preview)
 - OpenID Connect
 - GitHub

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<https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-app-service/5-authentication-authorization-app-service>

Discover App Service networking features

Multitenant App Service networking features

- Inbound features
 - App-assigned address
 - Access restrictions
 - Service endpoints
 - Private endpoints
- Outbound features
 - Hybrid Connections
 - Gateway-required VNet Integration
 - VNet Integration

Single-tenant networking

Azure App Service Environment hosts Isolated SKU plans directly in your Azure virtual network.

- External: Exposes the hosted apps by using an IP address that is accessible on the internet.
- Internal load balancer: Exposes the hosted apps on an IP address inside your virtual network.

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<https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-app-service/6-network-features>

- Regional virtual network integration: When you connect to virtual networks in the same region, you must have a dedicated subnet in the virtual network you're integrating with.
- Gateway-required virtual network integration: When you connect directly to virtual networks in other regions or to a classic virtual network in the same region, you need an Azure Virtual Network gateway created in the target virtual network.

Exercise: Create a static HTML web app by using Azure Cloud Shell

- In this exercise, you deploy a basic HTML+CSS site to Azure App Service by using the Azure CLI `az webapp up` command.
- You then update the code and redeploy it by using the same command.

Objectives

- Download the sample app
- Create the web app
- Update and redeploy the app

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Students can perform this exercise in their lab environment. Step-by-step instructions are included in the student manual.

<https://learn.microsoft.com/en-us/training/modules/introduction-to-azure-app-service/7-create-html-web-app>

Summary and knowledge check

In this module, you learned how to:

- Describe Azure App Service key components and value.
- Explain how Azure App Service manages authentication and authorization.
- Identify methods to control inbound and outbound traffic to your web app.
- Deploy an app to App Service using Azure CLI commands.

- 1 Which App Service plan category provides the maximum scale-out capabilities?
- 2 What networking feature of App Service can be used to control outbound network traffic?

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1. The Isolated category provides network and compute isolation, and has the maximum scale-out capability
2. Hybrid Connections are an outbound network feature.

Module 2: Configure web app settings

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<https://learn.microsoft.com/training/modules/configure-web-app-settings/>

Learning objectives

- Create application settings that are bound to deployment slots.
- Explain the options for installing TLS certificates for your app.
- Enable diagnostic logging for your app to aid in monitoring and debugging.
- Create virtual app to directory mappings.

Introduction

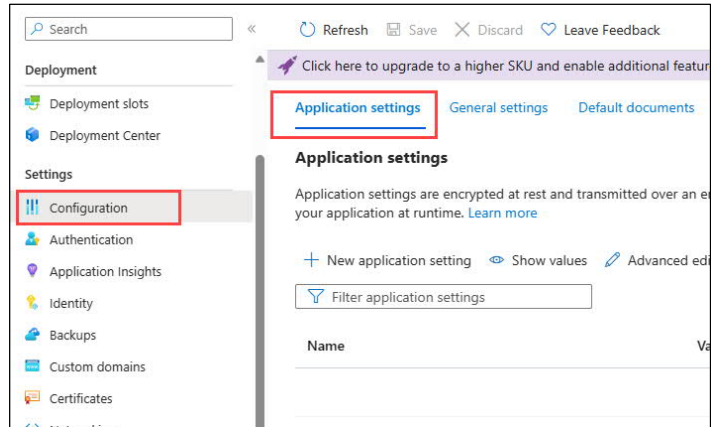
- In App Service, app settings are passed as environment variables to the application code.
- For Linux apps and custom containers, App Service passes app settings to the container using the `--env` flag to set the environment variable in the container.

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<https://learn.microsoft.com/en-us/training/modules/configure-web-app-settings/2-configure-application-settings>

Configure application settings (1 of 2)

- Adding and editing settings
 - To add a new app setting, click New application setting.
 - To add or edit app settings in bulk, click the Advanced edit button.
- Configure connection strings
 - Adding and editing connection strings follow the same principles as other app settings and they can also be tied to deployment slots.



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- For ASP.NET and ASP.NET Core developers, setting app settings in App Service are like setting them in `<appSettings>` in `*Web.config*` or `*appsettings.json*`, but the values in App Service override the ones in `*Web.config*` or `*appsettings.json*`.
- You can keep development settings (for example, local MySQL password) in `*Web.config*` or `*appsettings.json*`, but production secrets (for example, Azure MySQL database password) safe in App Service.

Configure application settings (2 of 2)

Editing settings in bulk

```
[
  {
    "name": "name-1",
    "value": "conn-string-1",
    "type": "SQLServer",
    "slotSetting": false
  },
  {
    "name": "name-2",
    "value": "conn-string-2",
    "type": "PostgreSQL",
    "slotSetting": false
  },
  ...
]
```

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You can add and edit settings in bulk by using the format shown above which illustrates setting connection strings.

Configure general settings

Stack settings

The software stack to run the app, including the language and SDK versions.

Platform settings

Configure settings for the hosting platform, including:

- WebSocket protocol
- Always On
- Managed pipeline version
- HTTP version
- ARR affinity

Debugging

Enable remote debugging for ASP.NET , ASP.NET Core, or Node.js apps.

Incoming client certificates

Require client certificates in mutual authentication. TLS mutual authentication is used to restrict access to your app by enabling different types of authentication for it.

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<https://learn.microsoft.com/en-us/training/modules/configure-web-app-settings/3-configure-general-settings>

Configure path mappings

Linux and containerized apps

- Add custom storage for your containerized app.
- Containerized apps include all Linux apps and Windows and Linux custom containers running on App Service.

Windows apps (uncontainerized)

- Customize the IIS handler mappings and virtual applications and directories.
- Handler mappings enable adding custom script processors to handle requests for specific file extensions.

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In the Configuration > Path mappings section you can configure handler mappings, and virtual application and directory mappings. The Path mappings page displays different options based on the OS type.

Windows apps (uncontainerized)

For Windows apps, you can customize the IIS handler mappings and virtual applications and directories.

Handler mappings let you add custom script processors to handle requests for specific file extensions. To add a custom handler, select New handler mapping. Configure the handler as follows:

- Extension: The file extension you want to handle, such as **.php* or *handler.fcgi*.
- Script processor: The absolute path of the script processor. Requests to files that match the file extension are processed by the script processor. Use the path *D:\home\site\wwwroot* to refer to your app's root directory.
- Arguments: Optional command-line arguments for the script processor.

Each app has the default root path (*/*) mapped to *D:\home\site\wwwroot*, where your code is deployed by default. If your app root is in a different folder, or if your repository has more than one application, you can edit or add virtual applications and directories.

You can configure virtual applications and directories by specifying each virtual directory and its corresponding physical path relative to the website root (D:\home). To mark a virtual directory as a web application, clear the Directory check box.

Linux and containerized apps

You can add custom storage for your containerized app. Containerized apps include all Linux apps and also the Windows and Linux custom containers running on App Service. Select New Azure Storage Mount and configure your custom storage as follows:

- Name: The display name.
- Configuration options: Basic or Advanced. Select Basic if the storage account isn't using service endpoints, private endpoints, or Azure Key Vault. Otherwise, select Advanced.
- Storage accounts: The storage account with the container you want.
- Storage type: Azure Blobs or Azure Files. Windows container apps only support Azure Files. Azure Blobs only supports read-only access.
- Storage container: For basic configuration, the container you want.
- Share name: For advanced configuration, the file share name.
- Access key: For advanced configuration, the access key.
- Mount path: The absolute path in your container to mount the custom storage.
- Deployment slot setting: When checked, the storage mount settings also apply to deployment slots.

Enable diagnostic logging

Type	Platform	Description
Application logging	Windows, Linux	Logs messages generated by your application code. The messages are generated by the web framework you choose.
Web server logging	Windows	Raw HTTP request data in the W3C extended log file format.
Detailed error messages	Windows	Copies of the <i>.html</i> error pages that would have been sent to the client browser.
Failed request tracing	Windows	Detailed tracing information on failed requests
Deployment logging	Windows, Linux	Deployment logging happens automatically and there are no configurable settings for deployment logging.

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<https://learn.microsoft.com/en-us/training/modules/configure-web-app-settings/5-enable-diagnostic-logging>

Configure security certificates

- Options for adding certificates in App Service
 - Free App Service managed certificate
 - Purchase an App Service certificate
 - Import a certificate from Key Vault
 - Upload a private certificate
 - Upload a public certificate

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<https://learn.microsoft.com/en-us/training/modules/configure-web-app-settings/6-configure-security-certificates>

A certificate uploaded into an app is stored in a deployment unit that is bound to the app service plan's resource group and region combination (internally called a webspace). This makes the certificate accessible to other apps in the same resource group and region combination.

Notes on certificates:

Free App Service managed certificate: A private certificate that's free of charge and easy to use if you just need to secure your custom domain in App Service.

Purchase an App Service certificate: A private certificate that's managed by Azure. It combines the simplicity of automated certificate management and the flexibility of renewal and export options.

Import a certificate from Key Vault: Useful if you use Azure Key Vault to manage your certificates.

Upload a private certificate: If you already have a private certificate from a third-party provider, you can upload it.

Upload a public certificate: Public certificates are not used to secure custom domains, but you can load them into your code if you need them to access remote resources.

Summary and knowledge check

In this module, you learned how to:

- Create application settings that are bound to deployment slots.
- Explain the options for installing SSL/TLS certificates for your app.
- Enable diagnostic logging for your app to aid in monitoring and debugging.
- Create virtual app to directory mappings.

1 Which app configuration settings category is used to set the language and SDK version?

2 What types of diagnostic logging are supported on the Linux platform?

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1. **General Settings** is used to configure stack, platform, debugging, and incoming client certificate settings.
2. Application and deployment logging are supported on the Linux platform.

Module 3: Scale apps in Azure App Service

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<https://learn.microsoft.com/training/modules/scale-apps-app-service/>

Learning objectives

- Identify scenarios for which autoscaling is an appropriate solution
- Create autoscaling rules for a web app
- Monitor the effects of autoscaling

Introduction

- Autoscaling adjusts the number of available instances to meet the varying demands on your application
- Create rules to specify the conditions where instances should be added or removed
- Control costs

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Examine autoscale factors

- Autoscaling adjusts available resources based on the current demand.
- Autoscaling performs scaling in and out, as opposed to scaling up and down.
- Monitors the resource metrics of a web app as it runs and detects when additional resources are required based on the set conditions.
- When should you consider autoscaling?
 - Autoscaling provides elasticity for your services.
 - Autoscaling improves availability and fault tolerance.
 - Autoscaling isn't the best approach to handling long-term growth.

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Autoscaling rules

Autoscaling makes its decisions based on rules that you define. A rule specifies the threshold for a metric, and triggers an autoscale event when this threshold is crossed. Autoscaling can also deallocate resources when the workload has diminished.

Per-app scaling can be enabled at the App Service plan level to allow for scaling an app independently from the App Service plan that hosts it. This way, an App Service plan can be scaled to 10 instances, but an app can be set to use only five. Per-app scaling is available only for Standard, Premium, Premium V2, Premium V3, and Isolated pricing tiers.

Autoscaling has an overhead associated with monitoring resources and determining whether to trigger a scaling event. In this scenario, if you can anticipate the rate of growth, manually scaling the system over time may be a more cost effective approach.

Identify autoscale factors

Autoscale conditions

- Scale based on a metric, such as CPU usage or the length of a disk queue.
- Scale based on a schedule, such as a time of day or day of the week.
- Can create multiple conditions to handle complex needs.

Autoscale metrics

- Metrics are based on all instances of an app
- CPU percentage
- Memory percentage
- Disk queue length – outstanding I/O requests
- HTTP queue length – number of client requests
- Data in – number of bytes received
- Data Out – number of bytes sent

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Enable autoscale in App Service

- **Enable autoscaling**
 - Not all pricing tiers support autoscaling
 - Some are only single instance or limited to manual scaling
- **Add scale conditions**
 - A default scale condition is created
 - Edit the default or create additional conditions
- **Create scale rules**
 - Conditions can contain one or more scale rules
 - Rules can be set based on calendar, metric, or both
- **Monitor autoscaling behavior**
 - View autoscale changes on the Run history chart
 - Tracks number of instances and which condition triggered the change

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Explore autoscale best practices

- Ensure the maximum and minimum values are different and have an adequate margin between them
- Choose the appropriate statistic for your diagnostics metric
- Choose the thresholds carefully for all metric types
- Check for conflicts when multiple rules are configured in a condition
- Always select a safe default instance count
- Configure autoscale notifications

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Summary and knowledge check

In this module, you learned how to:

- Identify scenarios for which autoscaling is an appropriate solution
- Create autoscaling rules for a web app
- Monitor the effects of autoscaling

- 1 How would you describe autoscaling?
- 2 Can you describe a scenario that is a suitable candidate for autoscaling?

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Autoscaling is a scale out/scale in solution. The system can scale out when specified resource metrics indicate increasing usage, and scale in when these metrics drop. The number of users requiring access to an application varies according to a regular schedule. For example, more users use the system on a Friday than other days of the week. Changes in application load that are predictable are good candidates for autoscaling.

Module 4: Explore Azure App Service deployment slots

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<https://learn.microsoft.com/training/modules/understand-app-service-deployment-slots/>

Learning objectives

- Describe the benefits of using deployment slots
- Understand how slot swapping operates in App Service
- Perform manual swaps and enable auto swap
- Route traffic manually and automatically

Introduction

- Deployment slots enable you to preview, manage, test, and deploy different development environments.
- Deployment slots are live apps with their own host names.
- App content and configuration elements can be swapped between two slots.

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<https://learn.microsoft.com/en-us/training/modules/understand-app-service-deployment-slots/>

Explore staging environments

- You can use a separate deployment slot instead of the default production slot when you're running in the Standard, Premium, or Isolated App Service plan tier.
- Deploying to a non-production slot has the following benefits:
 - You can validate app changes in a staging deployment slot before swapping it with the production slot.
 - Deploying an app to a slot first and swapping it into production makes sure that all instances of the slot are warmed up before being swapped into production.
 - After a swap, the slot with previously staged app now has the previous production app.

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Examine slot swapping

When swapping slots, App Service does the following

- 1 Applies the following settings from the target slot to all instances of the source slot:
 - Slot-specific app settings and connection strings, if applicable.
 - Continuous deployment settings, if enabled.
 - App Service authentication settings, if enabled.
- 2 Wait for every instance in the source slot to complete its restart.
- 3 If local cache is enabled, trigger local cache initialization by making an HTTP request to the application root ("/") on each instance of the source slot.
- 4 If auto swap is enabled with custom warm-up, trigger Application Initiation by making an HTTP request to the application root ("/") on each instance of the source slot.
- 5 If all instances on the source slot are warmed up successfully, swap the two slots by switching the routing rules for the two slots.
- 6 Now that the source slot has the pre-swap app previously in the target slot, perform the same operation by applying all settings and restarting the instances.

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Swap deployment slots

- Swap deployment slots on your app's Deployment slots page and the Overview page.
- Configure auto swap
- Swap with preview
- Specify a custom warm-up
- Roll back and monitor a swap

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Route traffic in App Service

Route production traffic automatically

- Go to your app's resource page and select Deployment slots.
- In the Traffic % column of the slot you want to route to, specify a percentage (between 0 and 100) to represent the amount of total traffic you want to route.

Route production traffic manually

- In addition to automatic traffic routing, App Service can route requests to a specific slot.
- This is useful when you want your users to be able to opt in to or opt out of your beta app.
- To route production traffic manually, you use the x-ms-routing-name query parameter.

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- By default, all client requests to the app's production URL (http://<app_name>.azurewebsites.net) are routed to the production slot.
- You can route a portion of the traffic to another slot.
- This feature is useful if you need user feedback for a new update, but you're not ready to release it to production.

Summary and knowledge check

In this module, you learned how to:

- Describe the benefits of using deployment slots
- Understand how slot swapping operates in App Service
- Perform manual swaps and enable auto swap
- Route traffic manually and automatically

- 1 What is the default routing rule applied to new deployment slots?
- 2 Do all configuration elements follow the content across a swap?

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1. By default, new slots are given a routing rule of 0%.
2. No, some elements (like WebJobs content) are swapped, and others (like WebJobs schedulers) are not.

End of presentation

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